

# Privacy Lens

## Evaluating Context-Aware Privacy Protection with Local LLMs

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### 1 Introduction

Consider an organisation that holds German free text about identifiable people, such as clinical intake notes, HR case files, or citizen correspondence. It may need to release or reuse this text for research, external services, or internal use without disclosing personal data. Article 9 of the GDPR places special categories of personal data, including health, ethnic origin, and religious beliefs, under stricter protection, making their detection and protection particularly important.

Named-entity redaction addresses only part of this problem. It can mask names, insurance numbers, and addresses, but it may leave untouched a sentence that contains no explicit disease term while still disclosing health information. For example, “Sara Hosseini spritzt sich seit der Diagnose vor jeder Mahlzeit Insulin” does not explicitly state a disease, but the combination of the diagnosis, insulin, and injection schedule reveals health information. Privacy protection for German text therefore requires two operations: removing direct identifiers through entity detection and determining whether sensitive information can be inferred from context.

Two further constraints shape a usable system. First, over-redaction has a cost: removing every clinically related term can protect privacy at the expense of the utility of the text. For example, a harmless sentence may contain similar medical vocabulary while referring to an animal rather than the person named in the sentence. Second, sensitive information is not always required to be redacted. Article 85 of the GDPR requires data protection to be balanced with freedom of expression and information, while Article 9(2)(e) provides an exception for data that the subject has manifestly made public. Consequently, a privacy-aware system must distinguish between private disclosures that require protection and information that is legitimately public in its context. This contextual judgement forms a central part of our evaluation.

We investigate whether locally hosted small language models can perform this task and whether task-specific fine-tuning improves their performance. Local execution is a requirement rather than a convenience: sending special-category personal data to a hosted API could itself constitute a privacy risk. Therefore, all systems evaluated in this work run locally on a single laptop.

We also examine whether models can distinguish private sensitive information from information that is legitimately public, such as disclosures made by public figures in relevant public or journalistic contexts. This tests whether models can apply privacy rules based on the surrounding context rather than simply masking sensitive words whenever they appear.

Our contributions are summarized as follows:

- First, we introduce three German datasets, including two evaluation sets based on minimal pairs that allow identifier removal and contextual privacy judgement to be evaluated separately.
- Second, we compare four prompted language models and a LoRA fine-tuned model against its base model under this evaluation protocol.

- Third, we show that fine-tuning substantially improves performance on explicit privacy detection while providing little or no improvement on contextual discrimination between sensitive and harmless sentences.

## 2 Related Work

Privacy-preserving NLP has traditionally relied on de-identification techniques such as Named Entity Recognition (NER) to detect and remove explicit personal information. Sousa and Kern [4] review these approaches and highlight the trade-off between privacy protection and the utility of processed text. Entity-based methods, however, may not capture information revealed indirectly through context, motivating the use of LLMs for contextual privacy risks. Agyemang et al. [1] evaluate eight privacy-preserving techniques and emphasize the importance of measuring information leakage after redaction. We extend this perspective to German-language data under GDPR requirements, considering both explicit identifiers and contextual special-category data.

Two further strands inform our design. Low-Rank Adaptation (LoRA) [2] freezes the pretrained model weights and trains small low-rank matrices, making task-specific adaptation feasible with limited computational resources. Minimal-pair evaluation, used for evaluating linguistic competence [5], pairs closely matched examples so that surface cues alone cannot determine the correct decision. We adopt this design for privacy evaluation to test whether models distinguish genuine privacy disclosures from harmless sentences with similar vocabulary.

## 3 Datasets

### 3.1 Dataset A

Dataset A is the main controlled evaluation set, containing 150 German sentences, of which 55 disclose an Article 9 attribute and 95 do not. Of the negative sentences, 55 form lexically matched twins with the positive sentences, resulting in 55 minimal pairs; the remaining 40 are standalone negatives. Both members of each pair share similar cue vocabulary but differ in whether a privacy disclosure is actually present, preventing systems from succeeding by relying on surface-level keywords alone.

The dataset covers five categories: implicit health (50 sentences), implicit ethnicity (40), mixed (20), explicit identifiers (20), and control sentences (20). Each sentence is additionally assigned a difficulty level. The dataset contains 340 gold spans, comprising 134 person names, 80 implicit cues that should be removed, 80 decoy cues that should be preserved, and 46 direct identifiers across ten types, including IBANs, tax numbers, and vehicle registration numbers.

### 3.2 Dataset B

Dataset B is the supervised training set used for LoRA fine-tuning. It contains 198 German examples in chat format, with 164 examples for training and 34 for development. Each example consists of a German system prompt, an original sentence, and a reference redaction. The prompt instructs the model to replace direct identifiers with placeholders and to rewrite sentences so that special-category information is no longer inferable. Dataset B is disjoint from the evaluation sets and uses reserved cue and attribute lexicons, ensuring that evaluation cues do not appear in the training data.

The training targets contain an important distinction between the two required privacy operations. Direct identifiers follow a regular transformation, with the same placeholder used across examples, whereas sensitive information is removed through context-dependent reformulation.

For example, “*Nach dem Herzinfarkt im Frühjahr übernimmt Ute Mahler keine Nachtschichten mehr*” is transformed into “*Aus gesundheitlichen Gründen übernimmt [NAME] keine Nachtschichten mehr*”. The identifier replacement is therefore highly consistent, while the rewriting of sensitive information varies across examples.

### 3.3 Dataset C

Dataset C evaluates whether a system can distinguish sensitive information that should be redacted from information that may legitimately be preserved. It contains 20 German sentences arranged into 9 case groups. Each group contrasts a private individual with a public office-holder and describes the same underlying fact, allowing the effect of the subject’s public role to be evaluated. Of the 20 items, 11 are labelled *REDACT* and 9 are labelled *PRESERVE*. Each preservation item is associated with a German rationale and a legitimacy basis: press freedom, self-disclosure, office relevance, public record, or constitutive relevance. The dataset covers seven Article 9 categories, with health and religious belief occurring most frequently.

**Data availability.** Dataset B is included in the repository, while Datasets A and C are not released. Instead, only their deterministic generators are provided to prevent models trained on the public repository from directly accessing the evaluation sentences and gold annotations. The generators use fixed seeds, so Dataset A can be reproduced exactly. Its released version is additionally identified by the MD5 checksum e6259a24dd877ff48d312ac10b2c25b5.

## 4 System Architecture

### 4.1 Prompted Pipeline

The four-model pipeline evaluates the four base LLMs under a common inference framework. Each model receives the same German evaluation sentences and the task-specific prompts described in Section 5. Inference is performed independently through Ollama, keeping privacy-sensitive text local.

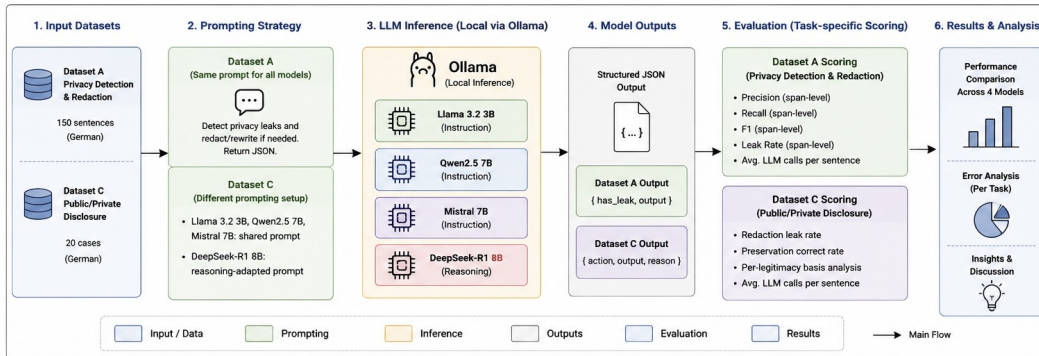


Figure 1: Architecture of the four-model privacy-processing pipeline.

The pipeline loads each sentence with a persistent identifier, combines it with the corresponding prompt, and sends it independently to each model. The models return structured JSON containing the privacy decision and transformed sentence. The identifier, model, configuration, output, and number of calls are stored in CSV files and passed to the offline evaluation scripts.

### 4.2 Fine-Tuning Pipeline

The second pipeline adapts Llama 3.2 3B to the privacy-processing task using LoRA and compares the resulting model with its untuned baseline. It uses the same evaluation sets and sealed-data protocol, but performs inference directly with the model weights rather than through Ollama and does not require a JSON output contract.

Dataset B is formatted using the model’s chat template, with the loss computed only on the assistant response. The pretrained weights remain frozen while the LoRA adapters are trained. Three fine-tuned variants use different random seeds, while the untuned model serves as the baseline. Outputs for Datasets A and C are evaluated offline using scoring scripts without loading model code.

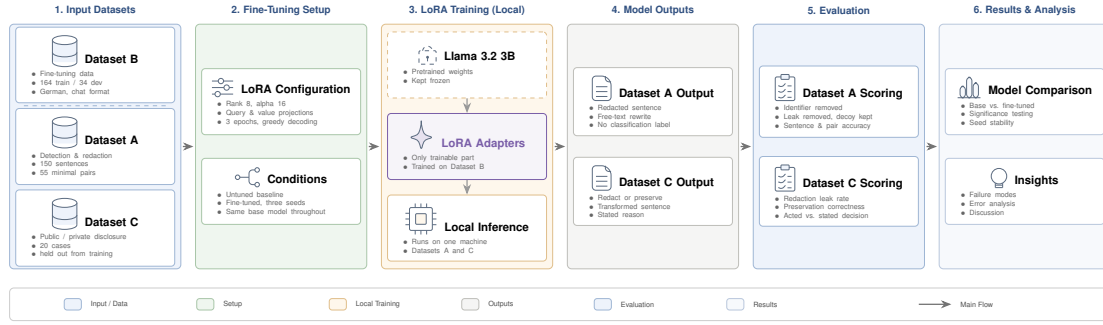


Figure 2: Architecture of the LoRA fine-tuning and evaluation pipeline.

## 5 Methods Compared

### 5.1 Prompted Models

We compare four locally hosted LLMs for privacy-sensitive German text processing. The selected models represent different model families, parameter sizes, and instruction-following characteristics:

Table 1: LLMs used in the four-model comparison.

| Model          | Parameters | Family   | Type        |
|----------------|------------|----------|-------------|
| Llama 3.2 3B   | 3B         | Llama    | Instruction |
| Qwen2.5 7B     | 7B         | Qwen     | Instruction |
| Mistral 7B     | 7B         | Mistral  | Instruction |
| DeepSeek-R1 8B | 8B         | DeepSeek | Reasoning   |

The models were selected to provide diversity in model family, parameter scale, and model type. Llama 3.2, Qwen2.5, and Mistral are instruction-tuned models, while DeepSeek-R1 represents a reasoning-oriented model. All four models were executed locally using Ollama, ensuring that the privacy-sensitive input data remained on the local machine and was not sent to external APIs. Predictions are evaluated using the task-specific scoring procedures described in Section 6.

### 5.2 Prompting Strategy

The models were prompted to perform privacy classification and text transformation directly on German input sentences. The prompts specified the privacy criteria, the required transformation, and the expected structured output.

For Dataset A, all four models used the same prompt. A privacy leak was defined as either an explicit identifier, such as a name, address, telephone number, or identification number, or sensitive information concerning the named person. The prompt distinguished information about the target person from information concerning other people, groups, animals, or professional activities. When a leak was detected, explicit identifiers were replaced with [MASKED], while sensitive information was removed or generalized. Unrelated information and the original German language were preserved.

For Dataset C, Llama 3.2, Qwen2.5, and Mistral used a shared privacy-action prompt. In addition to direct identifiers and sensitive information, the prompt introduced legitimate public disclosure as an exception. The models therefore selected one of four actions: `none`, `mask`, `rewrite`, or `mask_and_rewrite`. The prompt also required a short English explanation of the decision.

DeepSeek-R1 used a separate prompt for Dataset C because of its reasoning-oriented generation behaviour. The prompt explicitly instructed the model not to expose reasoning and to return only the required JSON object. The same four privacy actions and public-disclosure criteria were retained, with the output consisting of the selected action, transformed German sentence, and a short English reason.

All prompts required valid JSON without Markdown, code fences, or additional text. The input and transformed sentence were required to remain in German, while the `reason` field for Dataset C was required to be in English.

### 5.3 Fine-Tuned Model

The fine-tuning experiment investigates whether task-specific adaptation improves Llama 3.2 3B Instruct. The model is fine-tuned on Dataset B using LoRA and evaluated on the held-out Datasets A and C. The untuned Llama 3.2 3B Instruct serves as its baseline, isolating the effect of fine-tuning from model choice.

Three fine-tuned variants are trained with different random seeds and evaluated using the same procedure as the baseline. The configuration is summarized in Table 5. The prompted and fine-tuning tracks use different output contracts and model configurations, so their results are reported separately rather than compared on shared plots.

## 6 Evaluation

Both tracks are evaluated on the same sentences, but their output contracts differ, so not all metrics apply to both. The prompted pipeline produces a privacy decision together with its rewrite, whereas the fine-tuned model produces only a rewrite. Detection metrics are therefore reported only for the prompted models and shown as dashes for the fine-tuned model.

**Dataset A.** For the prompted track, we report precision, recall, and F1 against the sensitive/non-sensitive gold labels, together with leak rate over sentences that are sensitive in the gold. False negatives count as leaks because the sensitive sentence remains unprotected. For the fine-tuned track, the rewrite is evaluated against the annotated spans using three criteria: **identifier removed**, no fragment of a person-name span survives; **leak removed**, no implicit cue survives on positive sentences; and **decoy kept**, every decoy cue survives on negative sentences. A sentence is correct only if all applicable criteria hold. Fourteen sentences have no applicable span and are excluded, leaving 136 sentences in both conditions, making the comparison paired.

**Pair accuracy.** The criteria are particularly informative when evaluated on minimal pairs. In “*Sara Hosseini spritzt sich seit der Diagnose vor jeder Mahlzeit Insulin*”, the implicit cues must be removed, whereas in its matched twin, “*Sara Hosseini spritzt der alten Katze . . . zweimal täglich Insulin*”, the same cues must be preserved because the subject is a cat. A pair is correct only if both members are handled correctly. Thus, always redacting or always preserving the shared cues cannot achieve high pair accuracy. We additionally report over-redactions, where a negative sentence loses a decoy cue.

**Dataset C.** *REDACT* items are evaluated using leak rate, the proportion of outputs in which the person’s name or Art. 9 attribute remains present. A *PRESERVE* item is correct only if the name remains unchanged, the attribute remains explicit, and no new fact is introduced. Results are also broken down by legitimacy basis. Eight conditions are compared: the four prompted models, and the baseline and fine-tuned model under both the default and legitimacy-aware prompts.

**Statistics.** Feeding the gold redactions through the scoring scripts yields 100% on all criteria. Proportions are reported with Wilson 95% confidence intervals [6]. The two fine-tuning conditions are compared using the exact McNemar test [3], since the same sentences and pairs are evaluated under both conditions. Zero-success cells are reported using exact one-sided binomial upper bounds.

## 7 Results

### 7.1 Dataset A: prompted models

Table 2: Dataset A results for the four prompted LLMs.

| Model          | Precision | Recall | F1    | Leak Rate |
|----------------|-----------|--------|-------|-----------|
| DeepSeek-R1 8B | 0.404     | 1.000  | 0.576 | 54.5%     |
| Llama 3.2 3B   | 0.458     | 0.407  | 0.431 | 83.3%     |
| Mistral 7B     | 0.429     | 0.491  | 0.458 | 81.8%     |
| Qwen2.5 7B     | 0.466     | 0.745  | 0.573 | 72.7%     |

All four models were evaluated with single-pass inference, one call per sentence. DeepSeek-R1 detects every sensitive sentence but at a precision of 0.404, flagging most negatives as well. Qwen2.5 reaches the best balance at F1 0.573. Precision sits between 0.40 and 0.47 for all four, close to the 0.367 positive rate of the dataset.

A classifier that simply answers positive on every sentence scores  $P = 0.367$ ,  $R = 1.000$ ,  $F1 = 0.537$ : llama3.2 and mistral fall below that floor, and the best model exceeds it by four points. Leak rates stay between 54.5% and 83.3%, so none of the models detects Art. 9 disclosure much better than a trivial baseline, and none reliably removes it once detected.

Leak rate here is computed by the prompted-track scorer, whose must-remove set includes person spans; the fine-tuning track in Section 7.2 uses a different one, so the two leak rates are not directly comparable. Five llama3.2 rows are missing because of invalid JSON, four of them ethnicity items, so that model is scored over 145 sentences. The other three models returned valid JSON on all 150, so this is a robustness failure specific to the 3B model and concentrated in one sensitive category rather than a property of the pipeline.

### 7.2 Dataset A: fine-tuning

Table 3: Dataset A results for the untuned Llama 3.2 3B and the LoRA fine-tuned model. Sentence accuracy is computed over 136 sentences, pair accuracy over 55 minimal pairs, identifier removal over 134 sentences, and the remaining criteria over 55 sentences each.

| Metric             | Base               | Fine-tuned                |
|--------------------|--------------------|---------------------------|
| Sentence accuracy  | 29.4% [22.4, 37.6] | <b>55.1%</b> [46.8, 63.3] |
| Pair accuracy      | 9.1% [3.9, 19.6]   | 5.5% [1.9, 14.9]          |
| Identifier removed | 44.8%              | <b>99.3%</b>              |
| Leak removed       | <b>56.4%</b>       | 12.7%                     |
| Decoy kept         | 34.5%              | <b>78.2%</b>              |
| Over-redactions    | 36                 | <b>12</b>                 |

Fine-tuning corrected 51 sentences and broke 16, an exact McNemar  $p = 2.2 \times 10^{-5}$ . At the pair level it gained 3 pairs and lost 5,  $p = 0.73$ . The aggregate metric moves; the discrimination metric does not, from the same 136 outputs; seeds 42, 1 and 2 give sentence accuracies of 55.1%, 54.4% and 55.1%, so run-to-run variance is negligible.

Two readings have to be resisted. The baseline’s higher leak removal is not evidence of better privacy judgement, but a by-product of indiscriminate rewriting: it damages 36 negatives against 12 and strips sensitive terms while destroying the sentence around them. The fine-tuned model fails in the opposite direction, removing the identifier cleanly while leaving the Art. 9 attribute untouched. This asymmetry is traceable to Dataset B, where identifier removal has one fixed surface form in 146 of 164 targets, whereas attribute removal has none. This is consistent with low-rank adaptation preferentially acquiring the operation with a consistent surface realisation, although what the adapter encodes is not directly observed.

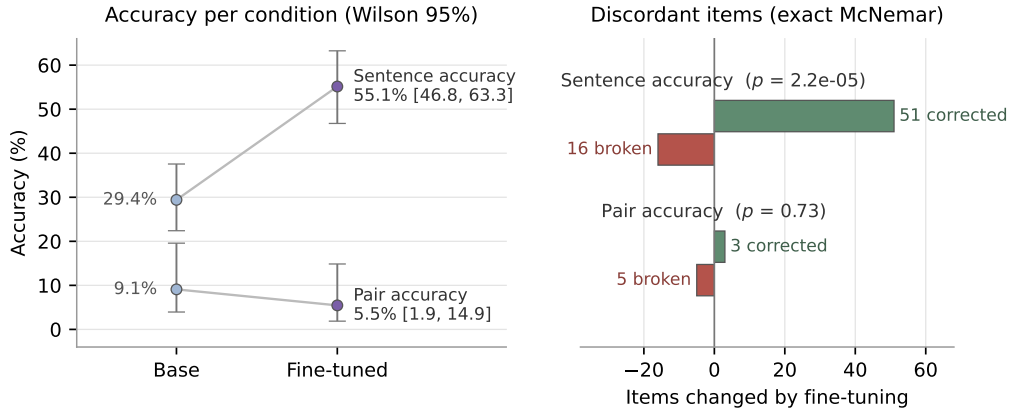


Figure 3: Effect of fine-tuning on Dataset A sentence and pair accuracy. Left: accuracy with Wilson 95% confidence intervals. Right: discordant cases used in the exact McNemar test.

### 7.3 Dataset C: all eight conditions

Table 4: Dataset C results for the four base LLMs. The preservation criterion does not separate a preserved attribute from a deleted one, as discussed below, so the PRESERVE figures are upper bounds.

| Model          | REDACT Leak Rate | PRESERVE Correct | Calls/Sent. |
|----------------|------------------|------------------|-------------|
| DeepSeek-R1 8B | 100.0%           | 11.1%            | 1.00        |
| Llama 3.2 3B   | 100.0%           | <b>22.2%</b>     | 1.00        |
| Mistral 7B     | 100.0%           | 0.0%             | 1.00        |
| Qwen2.5 7B     | <b>72.7%</b>     | 11.1%            | 1.00        |

Redaction leaks on almost every gold *REDACT* item across both tracks: three of the four prompted models leak on all of them, Qwen2.5 on 72.7%, and the fine-tuning conditions on 90.9% to 100%. Preservation is the harder half and neither track solves it. Among the prompted models Llama 3.2 scores highest at 22.2%, preserving two of three press-freedom cases, and DeepSeek-R1 one of three self-disclosed cases; no other legitimacy basis is handled correctly by any system.

The fine-tuning conditions (Table 6, appendix) act *REDACT* on 20 of 20 items in every one of the four, including the two whose prompt names Art. 85 and Art. 9(2)(e) explicitly, so a constant-redaction policy scores exactly the 11 gold *REDACT* items and nothing else, and pair accuracy is 0/7. Naming the statutory exemptions changed the acted decision on zero items. The one fine-tuning row scored as a preservation success is a deletion.

Given ‘*Ministerpräsident Anselm Kortner besuchte mit seiner Familie den Gottesdienst zum katholischen Erntedankfest im Dom*’ the output is that sentence with *zum katholischen Erntedankfest* removed, so the Art. 9 attribute is gone; it passes only because the name survived and the criterion cannot separate deletion from preservation. Discounting it leaves 0/9 in all four conditions, and the exact one-sided binomial upper bound for zero successes in nine puts preservation at no more than about 28% under any condition tested. With 20 sentences these results are a descriptive probe rather than a powered evaluation.

Finally, the second-pass justification contradicts the model’s own edit on 9 of 20 items for the baseline and 7 of 20 for the fine-tuned model, and re-eliciting it in English changes the stated decision on 8 and 7 of 20 respectively under identical weights and greedy decoding, so it is not a readout of the process that generated the edit.

## 8 Discussion

**Aggregate accuracy conceals the operation that matters.** Fine-tuning substantially improves sentence-level accuracy, but pair accuracy remains low. The model therefore becomes better at redaction without becoming better at distinguishing genuine disclosures from lexically matched harmless sentences.

**Transfer follows surface regularity.** Identifier removal uses a fixed [NAME] token in 146 of 164 training targets, whereas attribute removal has no canonical form. The near-complete transfer of identifier masking and the poor transfer of attribute removal suggest that fine-tuning favours operations with regular surface realisations.

**Contextual privacy remains unresolved.** Dataset C shows that models struggle to distinguish private disclosures requiring redaction from information that may legitimately remain public. The fine-tuned models redact all 20 items, even when the prompt explicitly states the relevant GDPR exceptions. Fine-tuning therefore improves redaction mechanics but not the contextual judgement of when redaction is appropriate.

**Scoring criteria encode assumptions.** The different scoring procedures in Datasets A and C highlight the need to evaluate privacy at the operation level. A unified evaluation should distinguish correctly removed information, leaked information, unnecessary deletion, and content-destroying rewriting rather than relying on aggregate accuracy alone.

## 9 Limitations

Model size is confounded with pipeline, since the fine-tuning track uses a 3B model throughout while the four-model comparison includes two 7B models and one 8B, so headline numbers are not compared across tracks. Detection metrics do not exist for the fine-tuning track, and inference cost is not comparable on Dataset C. Leak measurement is a string test, so a paraphrase that drops the token while keeping the meaning counts as protection and every rate is an upper bound; faithfulness is not scored, and no rule-based or majority-class baseline calibrates the absolute level. Effective sample size is below the nominal size, and all three datasets are synthetic, template-generated in one German register, covering two special categories in Dataset A and seven in Dataset C.

## 10 Conclusion

LoRA adaptation of Llama 3.2 3B substantially improved identifier removal but degraded implicit-attribute removal, leaving paired discrimination largely unchanged. The asymmetry is reflected in the training data: identifier removal has a fixed surface form, whereas attribute removal does not. On Dataset C, no condition reliably preserved legitimate public disclosures, and explicitly mentioning the relevant statutory exceptions did not change the model’s decisions.

The main contribution is therefore the evaluation design, which separates redaction from contextual privacy judgement rather than relying on aggregate accuracy alone.

Three follow-up experiments would test the findings further: normalising the Dataset B targets to give attribute removal a fixed surface form, adding identity, delete-all, and rule-based baselines, and evaluating the prompted, untuned, and fine-tuned Llama 3.2 3B under a common scorer.



## References

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## A Additional Results and Configuration

### A.1 LoRA Fine-Tuning Configuration

The configuration used for the LoRA experiments is shown in Table 5.

Table 5: LoRA fine-tuning configuration. Training takes approximately four minutes on an Apple M4 Max, with a final development loss of 0.330.

| Setting              | Value                 |
|----------------------|-----------------------|
| Base model           | Llama 3.2 3B Instruct |
| Adapters             | q_proj, v_proj        |
| Rank                 | $r = 8$               |
| LoRA $\alpha$        | 16                    |
| Dropout              | 0.05                  |
| Trainable parameters | 2.29M (0.07%)         |
| Learning rate        | $1 \times 10^{-4}$    |
| Scheduler            | Cosine                |
| Epochs               | 3                     |
| Batch size           | 8                     |
| Precision            | bfloat16              |
| Decoding             | Greedy                |
| Seeds                | {42, 1, 2}            |

### A.2 Dataset C Fine-Tuning Results

Table 6 reports the Dataset C results for the fine-tuning track under the default and legitimacy-aware prompts. The PRESERVE values are the raw scorer results and should be interpreted as upper bounds because the criterion can count deletion of the sensitive attribute as a preservation success.

Table 6: Dataset C results for the fine-tuning track under the default and legitimacy-aware prompts.

| Model | Prompt           | REDACT leak | PRESERVE |
|-------|------------------|-------------|----------|
| Base  | Default          | 100.0%      | 0/9      |
| Base  | Legitimacy-aware | 90.9%       | 1/9      |
| LoRA  | Default          | 90.9%       | 0/9      |
| LoRA  | Legitimacy-aware | 90.9%       | 0/9      |

### A.3 Application Figures

The following figures provide additional views generated by the PrivacyLens application. They complement the aggregate results in Section 7 by showing the trade-off between protection and preservation and the behaviour of systems on individual minimal pairs.

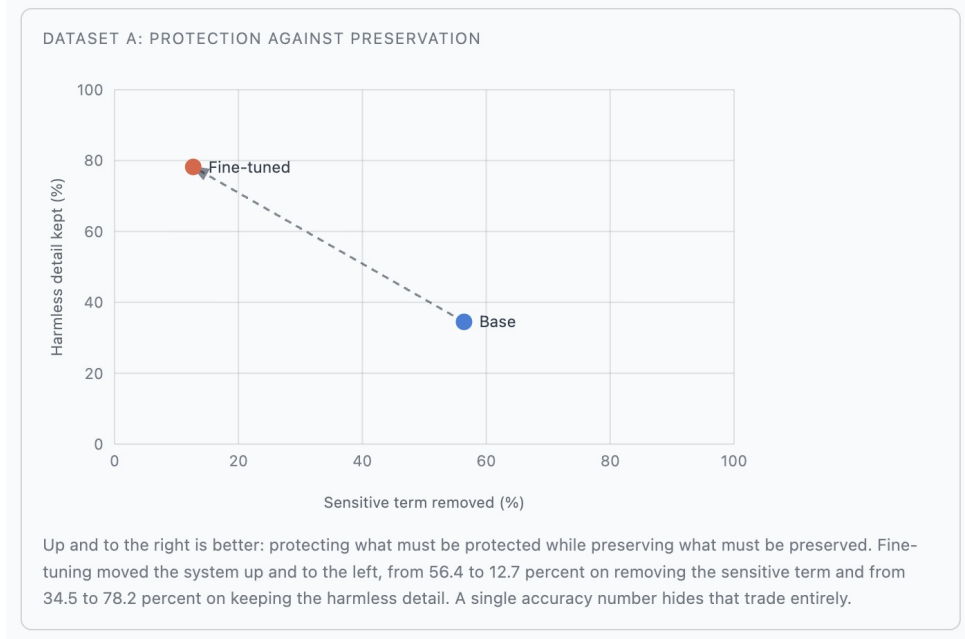


Figure 4: Trade-off between privacy protection and preservation. Each point represents one system and shows leak removal on positive items against decoy retention on negative items. The two tracks are shown separately because their scoring procedures differ.

Figure 4 makes the privacy–utility trade-off visible at the system level. The desirable region is the upper-right, where a system removes sensitive information while preserving harmless content. The figure shows that the systems do not simultaneously achieve both objectives, which is obscured by reporting a single aggregate accuracy value.

**Positive sentence.** This one really does disclose protected data (health). Correct behaviour: remove the name and remove or generalise the sensitive detail.  
Leaving the sensitive term in place is a leak, and the person is not protected.

GERMAN SENTENCE  
Sara Hosseini spritzt sich seit der Diagnose vor jeder Mahlzeit Insulin.

BASE MODEL, NO FINE-TUNING (Llama 3.2 3B) wrong name survived sensitive term survived  
INPUT, REMOVED MARKED  
Sara Hosseini ~~spritzt sich~~ seit ~~der Diagnose vor jeder Mahlzeit~~ Insulin.  
OUTPUT, ADDED MARKED  
Eine Person namens Sara Hosseini benötigt regelmäßig Insulin, seit sie eine bestimmte Erkrankung hat.  
GOLD TARGET  
[NAME] [SENSITIVE] sich seit der Diagnose vor jeder Mahlzeit [SENSITIVE].  
8 of 11 words removed or replaced (73 percent)

FINE-TUNED MODEL (LORA) (Llama 3.2 3B) wrong sensitive term survived  
INPUT, REMOVED MARKED  
~~Sara Hosseini~~ spritzt sich seit der Diagnose vor jeder Mahlzeit Insulin.  
OUTPUT, ADDED MARKED  
[NAME] spritzt sich seit der Diagnose vor jeder Mahlzeit Insulin.  
GOLD TARGET  
[NAME] [SENSITIVE] sich seit der Diagnose vor jeder Mahlzeit [SENSITIVE].  
2 of 11 words removed or replaced (18 percent)

**Negative sentence.** Same vocabulary, but the subject is an animal, not a person, so nothing here is protected. Correct behaviour: remove the name and change nothing else.  
Deleting the harmless detail here is over-redaction: information was destroyed for no reason.

GERMAN SENTENCE  
Sara Hosseini spritzt der alten Katze seit dem Befund des Tierarztes zweimal täglich Insulin.

BASE MODEL, NO FINE-TUNING (Llama 3.2 3B) wrong name survived harmless detail destroyed  
INPUT, REMOVED MARKED  
Sara Hosseini ~~spritzt~~ der alten Katze seit dem ~~Befund des Tierarztes zweimal täglich~~ Insulin. Sara Hosseini behandelt der alten Katze seit dem Arztbesuch regelmäßig Insulin.  
OUTPUT, ADDED MARKED  
GOLD TARGET  
[NAME] spritzt der alten Katze seit dem Befund des Tierarztes zweimal täglich Insulin.  
6 of 14 words removed or replaced (43 percent)

FINE-TUNED MODEL (LORA) (Llama 3.2 3B) correct  
INPUT, REMOVED MARKED  
~~Sara Hosseini~~ spritzt der alten Katze seit dem Befund des Tierarztes zweimal täglich Insulin. [NAME] spritzt der alten Katze seit dem Befund des Tierarztes zweimal täglich Insulin.  
OUTPUT, ADDED MARKED  
GOLD TARGET  
[NAME] spritzt der alten Katze seit dem Befund des Tierarztes zweimal täglich Insulin.  
2 of 14 words removed or replaced (14 percent)

Figure 5: Paired view of a positive sentence and its lexically matched twin, showing per-criterion verdicts and the resulting pair-level decision.

Figure 5 illustrates why pair accuracy is important. The two sentences share similar surface cues but require different privacy actions. A system that simply removes the shared sensitive-looking terms can therefore obtain good aggregate redaction scores while failing the actual contextual distinction. The pair view exposes which member of a minimal pair caused the failure and whether the error resulted from leakage or over-redaction.